1. PROJECT OVERVIEW

Robust and Explainable Classical Machine Learning for Tomato Disease Recognition under Real-World Field Conditions.

We are using images of tomato leaves belonging to six classes:

1. Healthy
2. Early blight
3. Late blight
4. Septoria leaf spot
5. Yellow leaf curl virus
6. Leaf mold

Instead of relying on deep learning models such as CNNs, we extract 105 HANDCRAFTED

FEATURES from each image.

These features describe things such as:

- COLOUR: What colours are present in the leaf?
- TEXTURE: What patterns and surface characteristics appear in the image?
- SHAPE: What geometric characteristics can be extracted?

We then use three classical machine learning models:

Model	Simple explanation
Random Forest (RF)	Combines many decision trees to make predictions.
Support Vector Machine (SVM)	Learns boundaries to distinguish between disease classes.
Logistic Regression (LR)	Learns relationships between features and classes using a linear model.

We want to answer four main questions:

- RQ1: How accurately can classical ML classify these six diseases?
- RQ2: How well do the models handle changes in image quality and lighting?

- RQ3: Can models obtain misleadingly high accuracy by learning similarities between related images or exploiting dataset shortcuts?
- RQ4: Which features are most useful, interpretable, and stable?

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2. What have we completed?

We have completed three major experimental stages.

Stage 1: Original dataset and baseline experiments

STATUS: COMPLETED.

We started with our original tomato leaf dataset.

What we did

- Inspected the dataset and its class distribution.
- Identified potential duplicate images and suspicious labels.
- Created a manifest to record image paths, classes, and dataset splits.
- Extracted 105 handcrafted features.
- Trained Random Forest, SVM, and Logistic Regression.
- Evaluated the models on ten different grouped splits.
- Tested how performance changed under 50 image perturbations, such as brightness changes, blur, shadows, and scaling.

A grouped split attempts to keep related images together rather than allowing related copies to appear in both training and testing.

What we found

The average macro-F1 scores using all features were:

Model	***Macro-F1***
Random Forest	0.600
SVM	0.615
Logistic Regression	0.596

MACRO-F1 measures classification performance across all six classes, giving each class equal importance.

Our baseline results showed that classical ML can recognise tomato diseases, but performance is not perfect.

We also discovered that:

- Blur and downscaling were particularly challenging.
- Some early-blight images were confused with Septoria.
- Certain lighting changes substantially affected predictions.
- Colour features were generally more useful than shape features in our experiments.
- Performance and feature importance could change under image perturbations.

We also investigated whether models could exploit similarities between images from particular sources.

IMPORTANT: We have baseline results, but not every question about feature importance, image shortcuts, and robustness has been conclusively answered.

Stage 2: Data augmentation experiments

STATUS: COMPLETED.

We wanted to know whether training on modified images could make the models more robust.

We compared three training conditions:

Condition	What we did
A0	Trained without additional augmentation
A1	Added geometric transformations, such as flips and rotations
A2	Added geometric transformations plus brightness and contrast changes

The additional augmented images were generated from training images only.

Validation and test images were not augmented for these experiments.

What we found

For Random Forest:

Condition	Clean macro-F1
A0	0.600
A1	0.612
A2	0.613

For SVM:

Condition	Clean macro-F1
A0	0.615
A1	0.633
A2	0.615

The clean-data improvements were not statistically significant after the multiple-comparison correction used in the experiment.

However, augmentation did improve performance under some conditions.

For example, A2 improved performance under certain brightness and contrast changes.

It also had trade-offs: some conditions, such as colour casts and noise, became worse for certain models.

What does this mean?

AUGMENTATION HELPED WITH SOME TYPES OF IMAGE VARIATION, BUT IT DID NOT CONSISTENTLY IMPROVE CLEAN-DATA PERFORMANCE OR ROBUSTNESS TO EVERY TYPE OF DISTORTION.

This is a useful finding because simply generating more images does not guarantee a better model.

Stage 3: Illumination normalisation experiments

STATUS: COMPLETED.

We tested two image-processing techniques intended to reduce the effects of lighting.

Technique	Simple explanation
Gray World	Adjusts image colours to make the average colour more neutral.
CLAHE	Enhances local contrast to make details more visible.

What we found

Model	Baseline macro-F1	Gray World	CLAHE
RF	0.600	0.548	0.611
SVM	0.615	0.553	0.612
LR	0.596	0.540	0.610

The scores are averages across the ten grouped splits.

Gray World reduced clean-data performance for all three models.

One reason was that changing the colours affected our leaf-segmentation process, which attempts to separate the leaf from the background.

CLAHE produced results broadly similar to the baseline, without a meaningful overall improvement.

What does this mean?

MORE IMAGE PREPROCESSING IS NOT NECESSARILY BETTER.

A technique may improve an image visually but still make classification worse if it changes the features our pipeline relies on.

This experiment also revealed an important limitation: the performance of an image-processing technique depends on the entire pipeline, including segmentation and feature extraction.

3. Augmented dataset experiment

STATUS: EXPERIMENT COMPLETED; DOCUMENTATION UPDATED, BUT THE FINAL DOCUMENTATION AUDIT REMAINS PENDING.

Your professor suggested using the supplied augmented dataset.

We inspected the dataset and ran our existing classical ML pipeline on it.

What we found in the dataset

The dataset contains 7,200 images:

Split	Images
Training	5,040
Validation	1,440
Testing	720
TOTAL	7,200

All six classes are balanced in each split.

We also discovered that many images have related versions across the training, validation, and test sets.

For example, a leaf image may appear in training, while a rotated or otherwise modified version of the same leaf appears in testing.

This matters because a model might recognise a familiar image rather than demonstrate that it can recognise a disease on a genuinely new leaf.

Our results on this dataset

Model	Test accuracy	Test macro-F1
Random Forest	81.0%	0.812
SVM	81.3%	0.815
Logistic Regression	69.6%	0.696

SVM achieved the highest numerical macro-F1, but its score was only 0.003 higher than Random Forest. We have not established that SVM is meaningfully better.

At first glance, these results look considerably better than our original grouped-split results.

However, WE MUST NOT CONCLUDE THAT OUR MODELS HAVE IMPROVED BY 18–21 PERCENTAGE POINTS. The datasets and evaluation procedures are different.

The most important discovery

We compared test images that had a detected related image in training or validation with those that did not.

Model	Accuracy with detected sibling	Accuracy without detected sibling
Random Forest	93.4%	68.5%
SVM	93.6%	68.8%
Logistic Regression	73.4%	65.7%

A *detected sibling* is an image identified as related to another image in training or validation through the filename or image-similarity rules we established.

Random Forest and SVM performed approximately 25 PERCENTAGE POINTS BETTER ON TEST IMAGES WITH A DETECTED SIBLING.

This is strong evidence that overlap between related images is associated with higher performance on the supplied split.

It does not prove that overlap explains every difference. The two groups have different class distributions, and our matching procedure cannot detect every possible related image.

What does this mean?

The supplied dataset gives us a benchmark score of approximately 0.81 macro-F1 for RF and SVM.

But that score does not tell us how accurately the models will recognise diseases on entirely new leaves in real-world conditions.

That distinction is central to our research.

4. Where do we stand against our four research questions?

Research question	Current status	What we know
RQ1: How well do classical ML models classify the six diseases?	SUBSTANTIALLY ANSWERED	We have baseline results on the original dataset and benchmark results on the supplied dataset.
RQ2: How robust are the models to image variations?	PARTIALLY ANSWERED	We tested many perturbations and augmentation methods. Performance depends on the type of variation and model.
RQ3: Can dataset shortcuts or related images inflate performance?	PARTIALLY ANSWERED	The supplied-split sibling analysis shows a strong association between detected overlap and higher accuracy. Some earlier source-related analyses also indicate potential shortcuts.
RQ4: Which features are useful and stable?	PARTIALLY ANSWERED	We have feature-importance and stability analyses, but some conclusions require careful verification and better documentation.

The project already has substantial experimental results. We do not need to start from zero.

However, some conclusions still need verification before we can confidently include them in the final paper.

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5. What is left to do?

There are several remaining tasks. They do not all have equal priority.

Task 1: Verify the project documentation

STATUS: PENDING. PRIORITY: HIGH.

We have updated three documents:

- `docs/verified_results.md`
- `docs/experiment_log.md`
- `docs/midterm_report_draft.md`

The updated documents record the supplied-dataset experiment and flag earlier statements that may be incorrect or overly strong.

For example, some older notes report different percentages for dataset overlap.

We need to check what those numbers actually measure and which evidence supports them.

We also need to review claims about:

- Whether related images were kept together in every experiment.
- How much shape features contribute.
- Whether the augmentation improvements came from geometric transformations or other changes.
- Whether previous conclusions about robustness are supported by the results.

WHAT REMAINS: Run a documentation consistency audit, resolve claims supported by evidence, and leave unresolved claims flagged.

This is the immediate next step.

Task 2: Run a grouped evaluation on the supplied dataset

Our current supplied-dataset experiment uses the dataset's existing split.

We have evidence that related images occur across those splits.

The next useful experiment would create new train/validation/test splits that keep detected related images together.

For example, if five images are versions of the same original leaf, we would try to keep all five in the same split.

We would then evaluate RF, SVM, and LR again.

Why is this important?

It would help answer:

DOES OUR MODEL STILL PERFORM WELL WHEN TEST IMAGES ARE LESS LIKELY TO HAVE RELATED COPIES IN TRAINING?

We should compare the new results with the supplied-split benchmark, while clearly labelling the different evaluation protocols.

We must also acknowledge that grouping methods cannot guarantee that every related image is detected.

This experiment would strengthen our evidence about the reliability of the supplied dataset's benchmark.

Task 3: Investigate the difficult disease classes

STATUS: PARTIALLY COMPLETED. PRIORITY: MEDIUM.

We have already observed confusion between:

- Early blight and Septoria.
- Yellow leaf curl and early blight.
- Yellow leaf curl and Septoria.

We should examine representative misclassified images and determine whether the errors are associated with similar visual symptoms, image quality, questionable labels, or other patterns.

We have also found cross-class near-duplicates in the supplied dataset.

We should investigate these cases rather than assuming that every label is correct.

However, we should not change labels without adequate evidence.

Task 4: Complete the literature survey

The midterm report needs a literature survey.

We need to review relevant research on:

- Classical machine learning for plant disease classification.
- Handcrafted colour, texture, and shape features.
- Robustness to lighting and image-quality changes.
- Data leakage and duplicate images in machine learning datasets.
- Explainability and feature-importance analysis.

The literature survey should explain what previous studies have done, what limitations remain, and how our project addresses those limitations.

We must use real, verifiable academic sources rather than inventing citations or claiming that our dataset has a published accuracy benchmark without evidence.

Task 5: Finish the midterm report

The main tasks are:

1. Complete the literature survey.
2. Finalise the problem statement and research questions.
3. Describe the dataset and methodology accurately.
4. Present the baseline, augmentation, illumination, and supplied-split results.
5. Explain the dataset-overlap issue and other limitations.
6. Describe the remaining work and mitigation strategies.
7. Verify that every numerical claim agrees with the experiment outputs.

The report must distinguish completed experiments from planned experiments.

We should not describe the grouped evaluation on the supplied dataset as completed until we actually run it.

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6. What about the original paper and its accuracy?

STATUS: DEFERRED.

We have not established a paper reporting benchmark accuracy for this exact supplied dataset.

The earlier investigation found dataset descriptions and licensing information but did not identify a linked paper with a reported accuracy.

Therefore, we have not reproduced a published benchmark.

That is fine for now. We can revisit the paper question later, as you suggested.

Our immediate priority is to understand and evaluate the dataset we have.

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7. What has been changed in our files?

The latest documentation update changed only these three files:

File	Purpose
`docs/verified_results.md`	Records verified experimental findings and limitations.
`docs/experiment_log.md`	Records what experiments were performed and what remains open.
`docs/midterm_report_draft.md`	Contains the report we are preparing for submission.

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8. Our deadlines and priorities

As of October 27, 2026:

Priority	Task	Status
1	Audit documentation and resolve conflicting claims	Pending
2	Complete literature survey	Pending
3	Refine and verify midterm report	In progress
4	Decide whether to run grouped evaluation on supplied dataset	Recommended; pending approval
5	Investigate important misclassifications and suspicious labels	Partially completed
6	Finalise research story and paper experiments	Pending
7	Finalise IEEE paper	Draft exists; needs further work

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9. What should we do next, in order?

I recommend this sequence:

FIRST: Audit the documentation. Make sure all numbers and conclusions are supported by the actual experimental evidence.

SECOND: Complete the literature survey. This is one of the largest outstanding report requirements.

THIRD: Update the midterm report using verified results from all completed experiments.

FOURTH: Decide whether to run the grouped evaluation on the supplied augmented dataset. This would be the most useful additional experiment for investigating the overlap problem.

FIFTH: Review misclassifications and refine the final paper's research contribution.

We should not begin more experiments just to generate more numbers. Every additional experiment should answer a specific research question.

The entire project in one minute

We built a classical ML pipeline using 105 handcrafted features and evaluated Random Forest, SVM, and Logistic Regression on tomato leaf disease images.

Our original grouped-split experiments achieved macro-F1 scores around 0.60–0.63. Augmentation improved robustness to some brightness and contrast changes, while Gray World illumination normalisation generally hurt performance and CLAHE had little overall effect.

On the supplied augmented dataset, RF and SVM achieved approximately 0.81 macro-F1. However, test accuracy was around 93% for images with detected training siblings and around 69% for images without detected siblings. This suggests that related-image overlap is an important concern when interpreting the supplied-split score.

We have documented these experiments, but we still need to audit conflicting claims, complete the literature survey, finish the midterm report, and decide whether to perform a grouped evaluation on the supplied dataset.